Abstract Writing

Title of the Reasearch Paper:

Advancements and Challenges in Quantum Computing: A Decade in Review

Purpose:

This review paper presents a comprehensive analysis of the major advancements and ongoing challenges in the field of quantum computing over the past decade. It aims to examine recent technological developments, evaluate research progress in quantum hardware and algorithms, and identify critical limitations that must be addressed for large-scale implementation.

Methodology:

The review incorporates peer-reviewed journal articles, technical reports, industry publications, and conference proceedings published between 2013 and 2023. The selected studies are analyzed based on advancements in qubit design, quantum error correction, algorithm development, and practical implementation models. Trends in research funding, experimental breakthroughs, and commercialization efforts are also evaluated.

Results:

The findings indicate significant improvements in quantum processor development, including increased qubit stability, enhanced coherence times, and more efficient quantum algorithms such as Shor’s and Grover’s algorithms. Major technology companies have demonstrated measurable progress in achieving quantum supremacy under controlled conditions. However, substantial challenges remain, including high error rates, scalability constraints, extreme cooling requirements near absolute zero (−273°C), and high infrastructure costs. Additionally, concerns regarding cybersecurity risks and the need for post-quantum cryptographic systems have gained increased research attention.

Conclusion:

The review concludes that quantum computing has transitioned from theoretical research to experimental validation, marking a critical phase in technological evolution. Despite remarkable progress, overcoming technical, economic, and operational challenges remains essential before widespread commercial adoption becomes feasible. Future research should focus on improving fault-tolerant architectures, scalable quantum systems, and secure quantum-resistant cryptographic frameworks.